

SRIMATHI SUNDARAVALLI MEMORIAL SCHOOL, CHENNAI 600 063
LAB PROGRAMS (16-20)
2026-2027

PGM NO: 16

SQL COMMAND – 1

Consider the following tables **FACULTY** and **COURSES**. Write SQL commands for the statements (i) to (vi) and give outputs for SQL queries (vii) to (x).

TABLE: FACULTY

F_ID	Fname	Lname	Hire_date	Salary
102	Amit	Mishra	12-10-1998	12000
103	Nitin	Vyas	24-12-1994	8000
104	Rakshit	Soni	18-5-2001	14000
105	Rashmi	Malhotra	11-9-2004	11000
106	Sulekha	Srivastava	5-6-2006	10000

TABLE: COURSES

C_ID	F_ID	Cname	Fees
C21	102	Grid Computing	40000
C22	106	System Design	16000
C23	104	Computer Security	8000
C24	106	Human Biology	15000
C25	102	Computer Network	20000
C26	105	Visual Basic	6000

GIVE QUERY:

- (i) To display details of those Faculties whose salary is greater than 12000.
- (ii) To display the details of courses whose fees is in the range of 15000 to 50000
(both values included).
- (iii) To increase the fees by 500 for “System Design” and “Human Biology” Courses.
- (iv) To display details of those courses which are taught by ‘Sulekha’ in descending order of courses name.
- (v) To display details of those Faculties whose First name starts with ‘R’.
- (vi) To display First Name, Last Name, Salary, Course Name and Fees from both the tables, whose salary is greater than 10000, Hire_date is above ‘01-01-2000’ and their Last name end with ‘a’.

GIVE OUTPUT:

- (vii) SELECT COUNT(DISTINCT F_ID) FROM COURSES;
- (viii) SELECT MIN(SALARY) FROM FACULTY,COURSES
WHERE COURSES.F_ID =FACULTY.F_ID;
- (ix) SELECT COUNT(*), MAX(HIRE_DATE), MIN(FNAME)
FROM FACULTY, COURSES
WHERE SALARY >= 12000 AND FEES < 50000 AND
COURSES.F_ID = FACULTY.F_ID;
- (x) SELECT * FROM FACULTY
WHERE HIRE_DATE BETWEEN '2001-01-01' AND '2005-12-31';

PGM.NO.: 17**SQL COMMAND – 2**

Study the following tables **CLIENT** and **BILL** and write SQL commands for the questions (i) to (vi) and give outputs for SQL queries (vii) to (x).

TABLE: CLIENT

CUST_ID	CUST_NAME	ADDRESS	PHONE_NO	CITY
C007	Pritam Sharma	12,M.G Road	71274250	Bangalore
C008	Sutopa	14/1 PritamPura	41206819	Delhi
C010	AnuragBasu	15A, Park Road	61281921	Kolkata
C012	Hrithik	7/2 VasantKunj	26121949	Delhi
C013	Firoz Shah	2, Servamali road	25014192	Bangalore
C025	Vinod Nagpal	46-a Navi Mumbai	64104944	Mumbai
C027	Sameer	41,Dwarka	42101619	Delhi
C002	Pasunjit Bose	16/A K.G Marg	27220012	Bangalore
C035	Aamina Begum	13/A Versova	41612181	Mumbai

TABLE: BILL

ORD_ID	CUST_ID	ITEM	ORD_DATE	QTY	PRICE
7002	C007	Pizza	20-NOV-07	1	249.50
7003	C013	Garlic Bread	24-OCT-05	3	75.75
7004	C012	Pasta	03-MAR-07	4	173.00
7005	C010	Ice Cream	01-JAN-08	30	195.75
7006	C035	Pizza	02-MAR-06	4	249.50
7009	C035	Garlic Bread	02-MAR-08	2	75.75
7010	C013	Brownie	04-MAY-07	4	40.50
7011	C014	Ice Cream	02-JUN-08	5	195.75
7012	C002	Pizza	01-FEB-08	7	249.50

GIVE QUERY:

- (i) To display a report containing cust_id, cust_name, Item, qty, price and bill amount. Bill amount is calculated as qty*price.
- (ii) To display how many customers have ordered Pizza in the month of March.
- (iii) To count the number of customer who have ordered item worth more than 1700.
Total amount = qty* price.
- (iv) To display the name of customer along with city in alphabetical order of city.
- (v) To display the count of each item from BILL table.
- (vi) To display the cust_name, address, item, ord_date for those quantities which are more than 3.

GIVE OUTPUT:

- (vii) SELECT CUST_NAME, CITY, ORD_DATE FROM CLIENT A, BILL B
WHERE A.CUST_ID = B.CUST_ID;
- (viii) SELECT CUST_NAME FROM CLIENT
WHERE CUST_ID IN(SELECT CUST_ID FROM BILL
WHERE (YEAR(ORD_DATE) ='2008'));
- (ix) SELECT CITY ,COUNT(*) FROM CLIENT
GROUP BY CITY;
- (x) SELECT DISTINCT (ITEM) FROM BILL;

Study the following tables **ITEM** and **CUSTOMER** and write SQL commands for the questions (i) to (vi) and give outputs for SQL queries (vii) to (x).

TABLE: ITEM

P_ID	ITEMNAME	MANUFACTURER	PRICE
PC01	Personal Computer	ABC	35000
LC05	Laptop	ABC	55000
PC03	Personal Computer	XYZ	32000
PC06	Personal Computer	COMP	37000
LC03	Laptop	PQR	57000

TABLE: CUSTOMER

C_ID	CUSTOMERNAME	CITY	P_ID
01	N.Roy	Delhi	LC03
06	H.Singh	Mumbai	PC03
12	R.Pandey	Delhi	PC06
15	C.Sharma	Delhi	LC03
16	K.Agarwalh	Bangalore	PC01

GIVE QUERY:

- (i) To display the details of those Customers whose city is Delhi.
- (ii) To display the details of Item whose Price is in the range of 35000 to 55000 (Both values included).
- (iii) To display the CustomerName, City from table Customer, and ItemName and Price from table Item, with their corresponding matching I_ID.
- (iv) To increase the Price of all Items by 1000 in the table Item.
- (v) To display the customer name and manufacturer of the highest priced item.
- (vi) To display the number of customers from each city.

GIVE OUTPUT:

- (vii) SELECT DISTINCT CITY FROM CUSTOMER;
- (viii) SELECT ITEMNAME, MAX(PRICE), COUNT(*) FROM ITEM
GROUP BY ITEMNAME;
- (ix) SELECT CUSTOMERNAME, MANUFACTURER FROM ITEM, CUSTOMER
WHERE ITEM.P_ID=CUSTOMER.P_ID
- (x) SELECT ITEMNAME, PRICE * 100 FROM ITEM
WHERE MANUFACTURER = 'ABC';

PGM.NO.: 19**SQL COMMAND – 4**

Consider the following tables **TEACHER** and **SALARY**. Write SQL commands for the statements (i) to (vii) and give outputs for SQL queries (viii) to (xii).

TABLE: TEACHER

SID	TNAME	DEPT	GENDER	DOB	EXPERIENCE
101	SIDDHARTH H	ACCOUNTS	M	1975-12-25	12
104	JOHN	PHYSICS	M	1993-01-13	5
107	NAMAN	CHEMISTRY	M	1994-08-12	4
114	NUPUR	ENGLISH	F	1978-07-09	15
109	JANVI	PHYSICS	F	1975-05-01	13
105	SELVA	ACCOUNTS	M	1964-06-21	24
117	JANAKI	COMPUTER	F	1967-03-17	18
111	BINOY	ACCOUNTS	F	1987-09-15	8
130	AMRESH	COMPUTER	M	1983-10-05	12

TABLE: SALARY

SID	BASIC	ALLOWANCE	DA
101	12000	1000	300
104	23000	2300	500
107	32000	4000	500
114	12000	5200	1000
109	42000	1700	200
105	18900	1690	300
130	21700	2600	300

GIVE QUERY:

- (i) To display the names and gender of all the staff who are in Accounts department and having more than 8 years of experience.
- (ii) To display the no of teachers, maximum and minimum BASIC in SALARY.
- (iii) To display the names and dept of all staff who doesn't have 'a' in their name.
- (iv) To display the SID and Department of the teachers in the descending order of their Experience.
- (v) To display the report containing SID, teacher's name, department, gender, Gross Salary (BASIC + ALLOWANCE + DA) and PF (BASIC*0.12).
- (vi) To display the details of the teachers who have their DOB in 1975 and Experience more than 12.

GIVE OUTPUT:

- (vii)

```
SELECT TNAME, DEPT
FROM TEACHER T, SALARY S
WHERE T.SID=S.SID AND 5*DA>ALLOWANCE;
```
- (viii)

```
SELECT * FROM TEACHER
WHERE DEPT IN ('English', 'Physics', 'Accounts') AND
DOB BETWEEN '1965-01-01' AND '1975-12-31';
```

(ix) SELECT SUM (BASIC+ALLOWANCE+DA) AS “TOTAL EXPENDITURE”,
 AVG (BASIC+ALLOWANCE+DA) AS “AVERAGE SALARY”
 FROM SALARY;

(x) SELECT DISTINCT (DEPT) FROM TEACHER;

(xi) SELECT TNAME ‘MOST EXPERIENCED TEACHER’
 FROM TEACHER
 WHERE EXPERIENCE = (SELECT MAX (EXPERIENCE) FROM TEACHER);

PGM.NO.: 20

SQL COMMAND – 5

Study the following tables **COMPANY** and **CUSTOMER** and write SQL commands for the questions (i) to (vi) and give outputs for SQL queries (vii) to (x).

TABLE: **COMPANY**

CID	NAME	CITY	PRODUCTNAME
111	SONY	DELHI	TV
222	NOKIA	MUMBAI	MOBILE
333	ONIDA	DELHI	TV
444	SONY	MUMBAI	MOBILE
555	BLACKBERRY	MADRAS	MOBILE
666	DELL	DELHI	LAPTOP

TABLE: **CUSTOMER**

CUSTID	NAME	PRICE	QTY	CID
101	ROHAN SHARMA	70000	20	222
102	DEEPAK KUMAR	50000	10	666
103	MOHAN KUMAR	30000	5	111
104	SAHIL BANSAL	35000	3	333
105	NEHA SONI	25000	7	444
106	SONAL AGGARWAL	20000	5	333
107	ARUN SINGH	50000	15	666

GIVE QUERY:

- (i) To display those company name which are having price less than 30000.
- (ii) To display the name of the companies in reverse alphabetical order.
- (iii) To increase the price by 1000 for those customer whose name starts with 'S'.
- (iv) To add one more column totalprice with decimal value to the table customer.
- (v) To display the details of company where productname as mobile.
- (vi) To display all the details from company and customer using natural join whose price value is above 40000.

GIVE OUTPUT:

- (vii)

```
SELECT S.CID, A.NAME, (PRICE*QTY)-(PRICE*0.02) AS GROSS
FROM COMPANY S, CUSTOMER A
WHERE S.CID = A.CID AND QTY > 5;
```
- (viii)

```
SELECT * FROM CUSTOMER
WHERE (NAME LIKE ' %A' OR NAME LIKE 'S%' ) AND PRICE <75000;
```
- (ix)

```
UPDATE CUSTOMER
SET QTY = QTY+1
WHERE CID IN(111,444,555,333)
```
- (x)

```
SELECT CID, NAME, CITY
FROM COMPANY
WHERE PRODUCTNAME = 'MOBILE';
```